

Fundamentals of Cyber Security and Digital Forensics

Module: 2

Priyanka Shingane

Module 2:

Lecture : Introduction to Cryptography



Cryptography

Cryptography *is the art and science of achieving security by encoding messages to make them non-readable.*

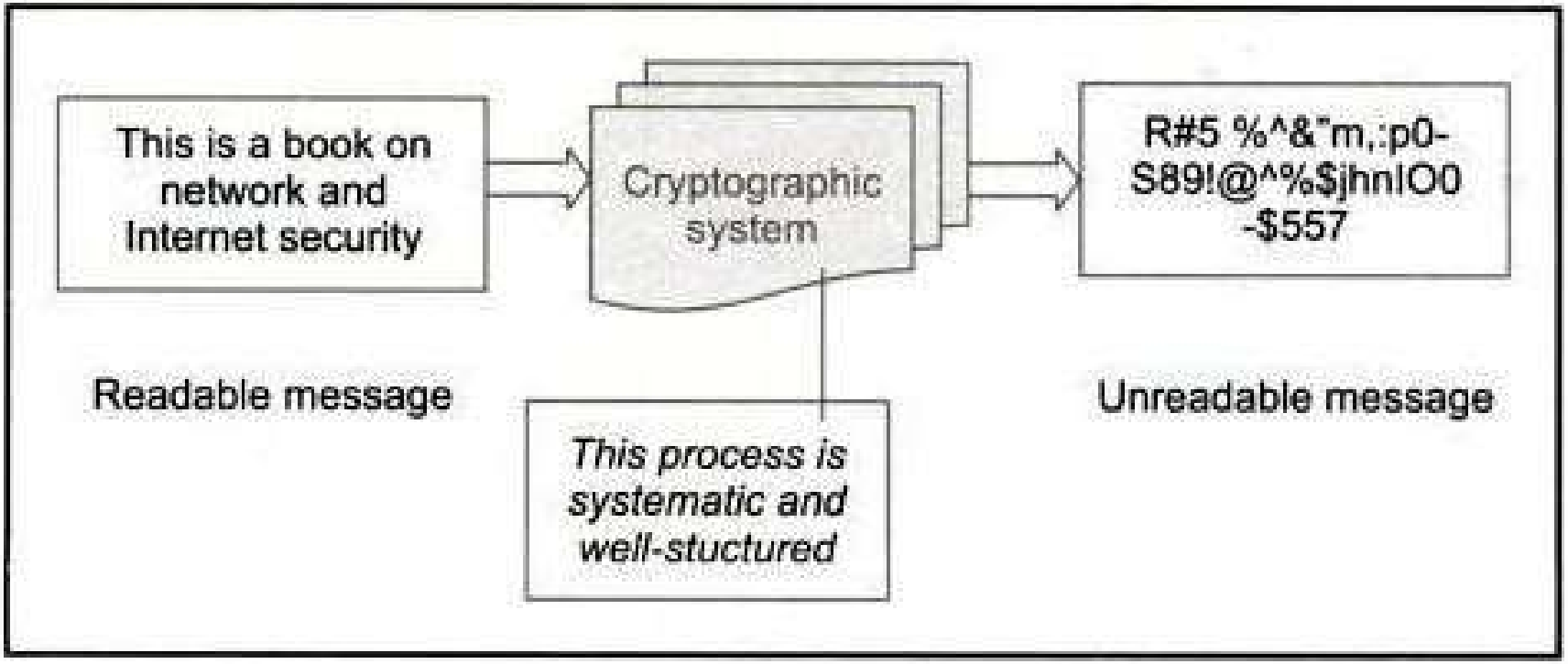


Fig. Cryptographic System

Cryptanalysis *is the technique of decoding messages from a non-readable format back to readable format without knowing how they were initially converted from readable format to non-readable format.*

Cryptography

Cryptology *is a combination of cryptography and cryptanalysis.*

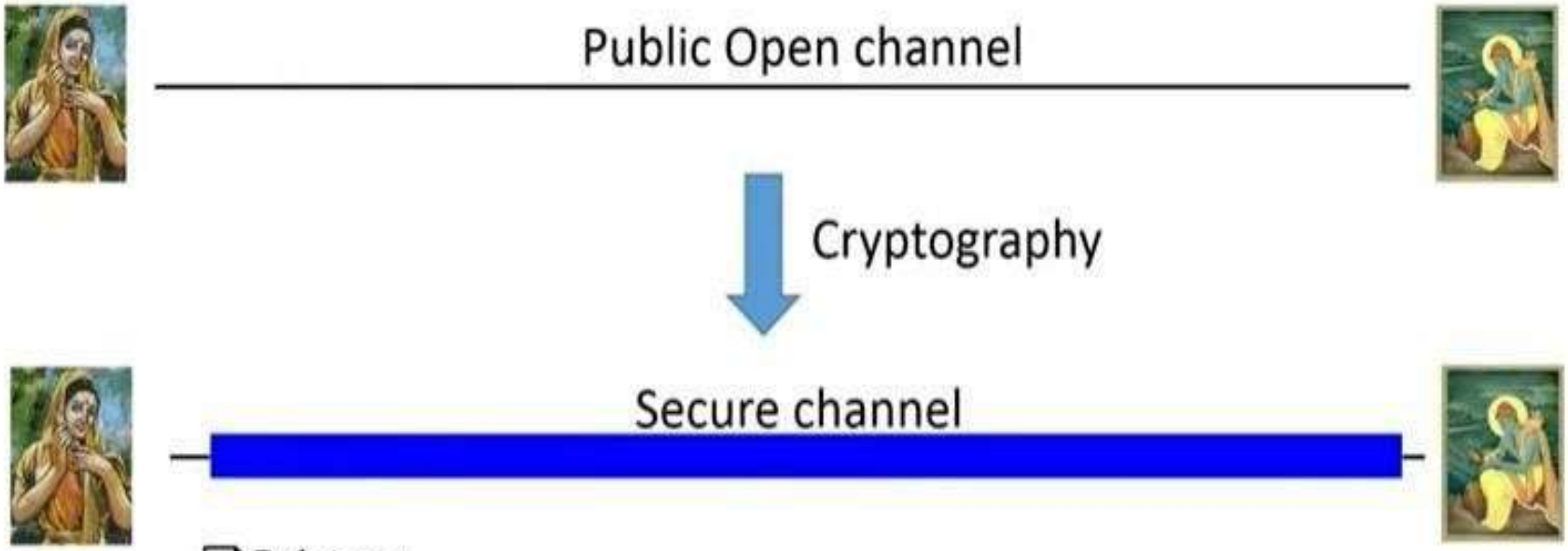
Plaintext :-

Clear text or plain text signifies a message that can be understood by the sender, the recipient and also by anyone else who gets an access to that message.

Ciphertext :-

When a plain text message is codified using any suitable scheme, the resulting message is called as cipher text.

Application of Cryptography:



- Privacy
- Authenticity
- Integrity

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Specific Applications:

- Secure Online Transactions:
- Secure Communication:
- Data Storage:
- Password Protection:
- Digital Signatures:
- Access Control:
- Blockchain Technology:
- Internet of Things (IoT):
- Virtual Private Networks (VPNs):
- Financial Transactions:
- Digital Currencies:
- Healthcare:
- Government and Defense:

CLASSICAL ENCRYPTION TECHNIQUES

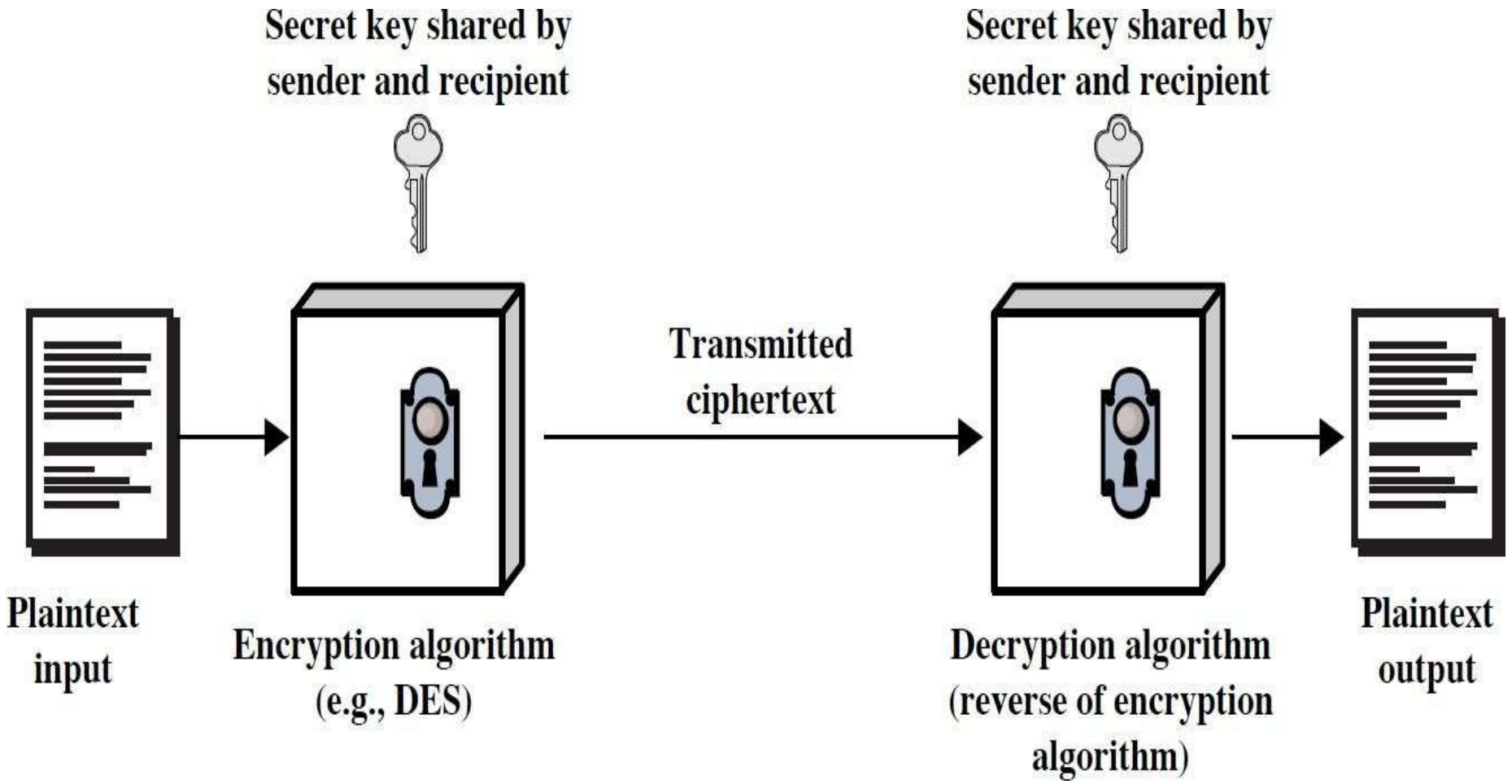
Basic Terminology:

- Plaintext: original message to be encrypted
- Ciphertext: the encrypted message
- Enciphering or encryption: the process of converting plaintext into ciphertext
- Encryption algorithm: performs encryption
  - Two inputs: a plaintext and key/keys



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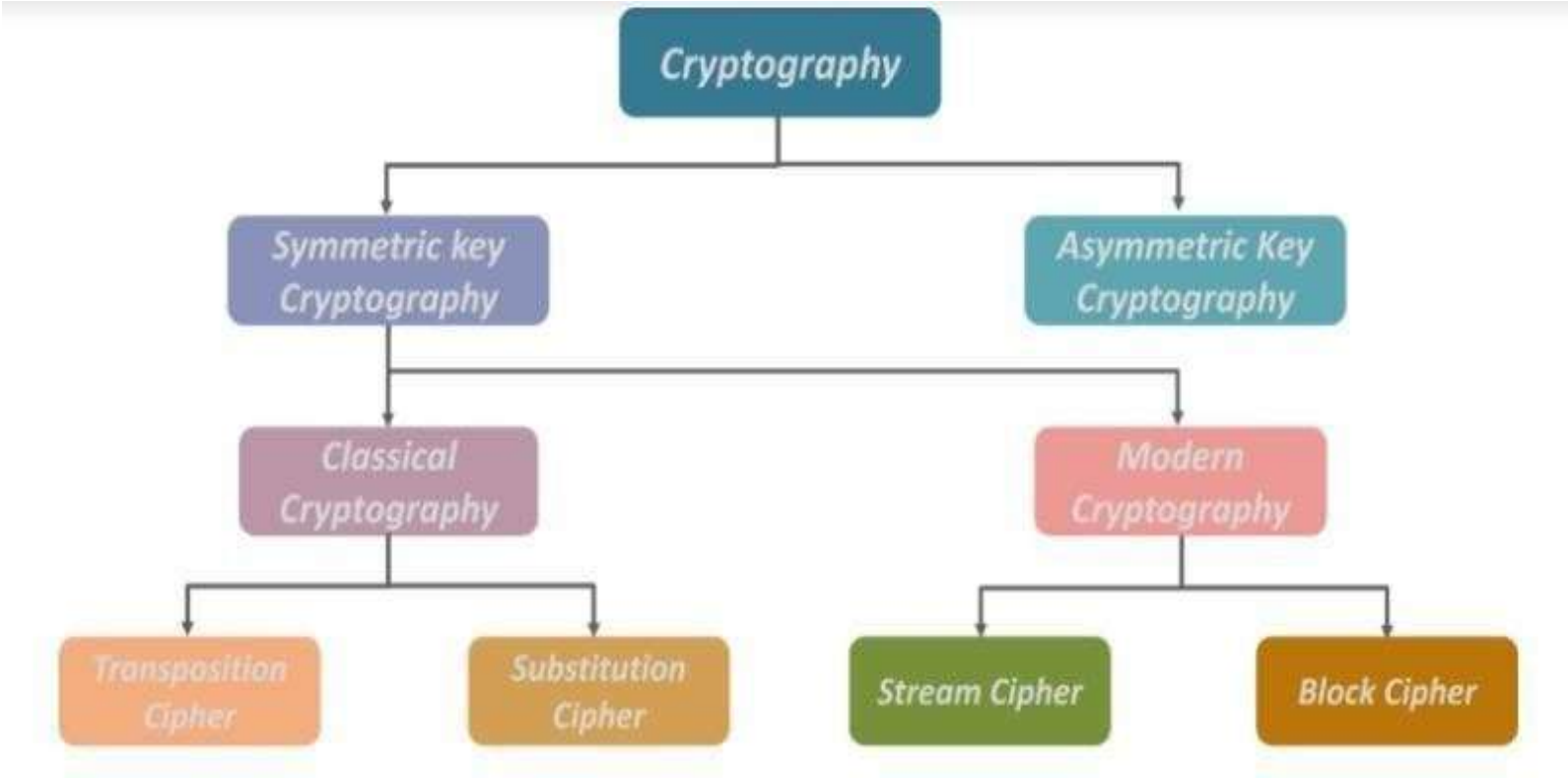
SYMMETRIC CIPHER MODEL



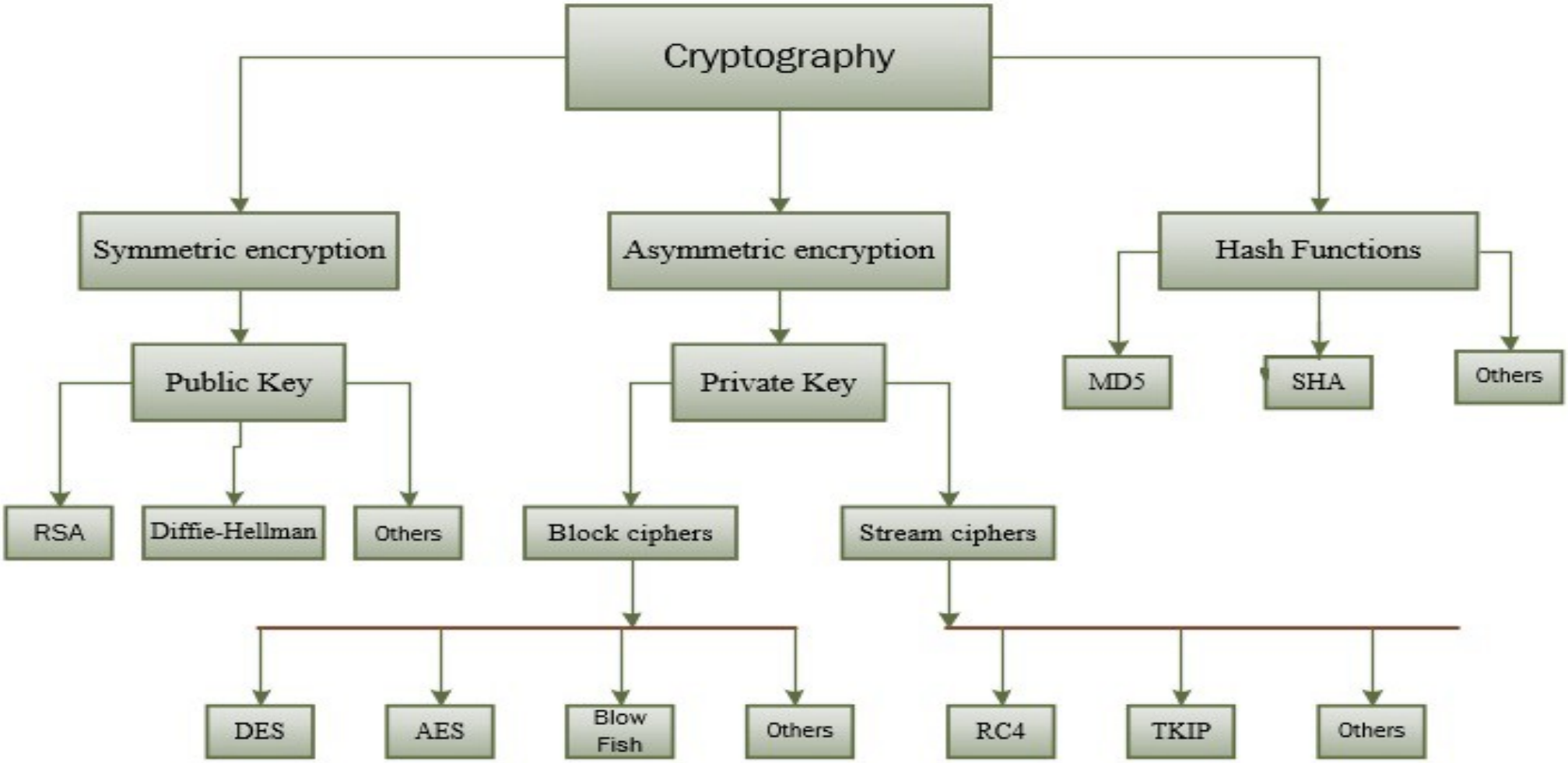
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CIPHERS

Symmetric cipher: same key used for encryption and decryption
Asymmetric cipher: different keys used for encryption and decryption
Block cipher: encrypts a block of plaintext at a time (typically 64 or 128 bits)
Stream cipher: encrypts data one bit or one byte at a time



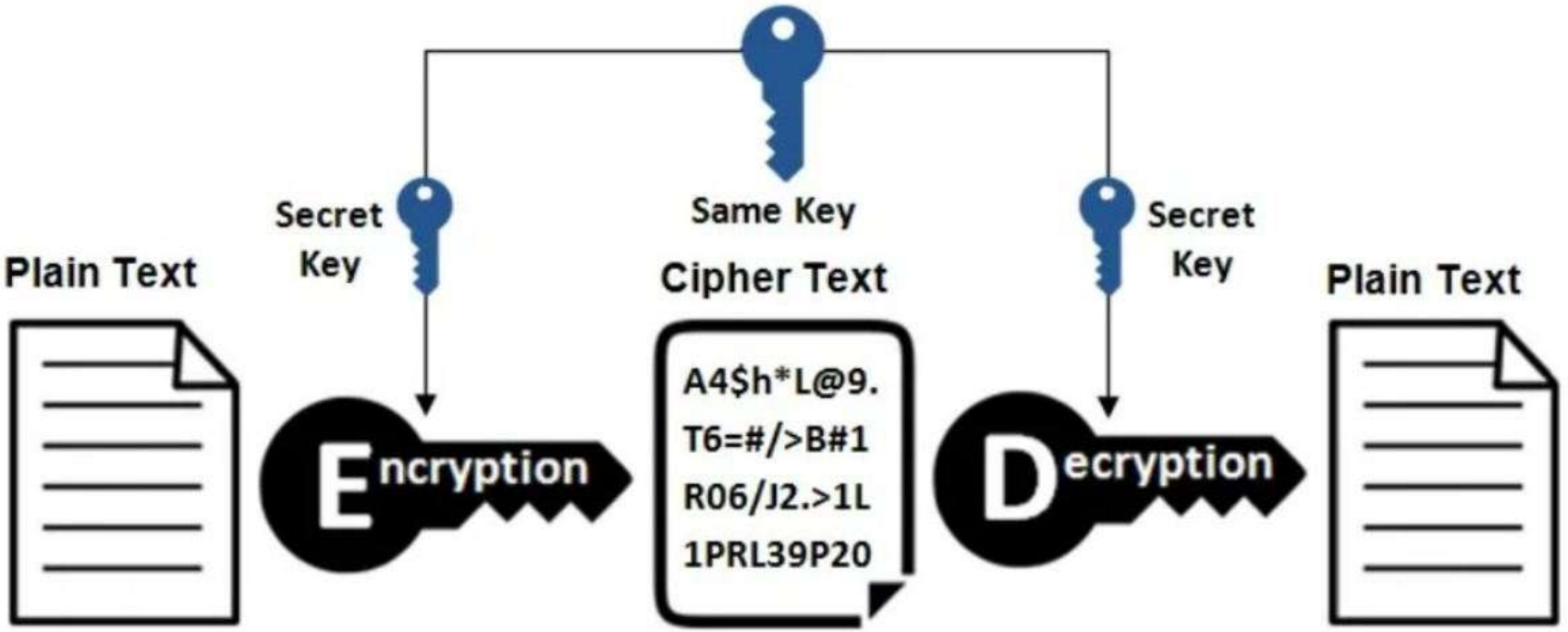
Cryptography is broadly classified into two categories



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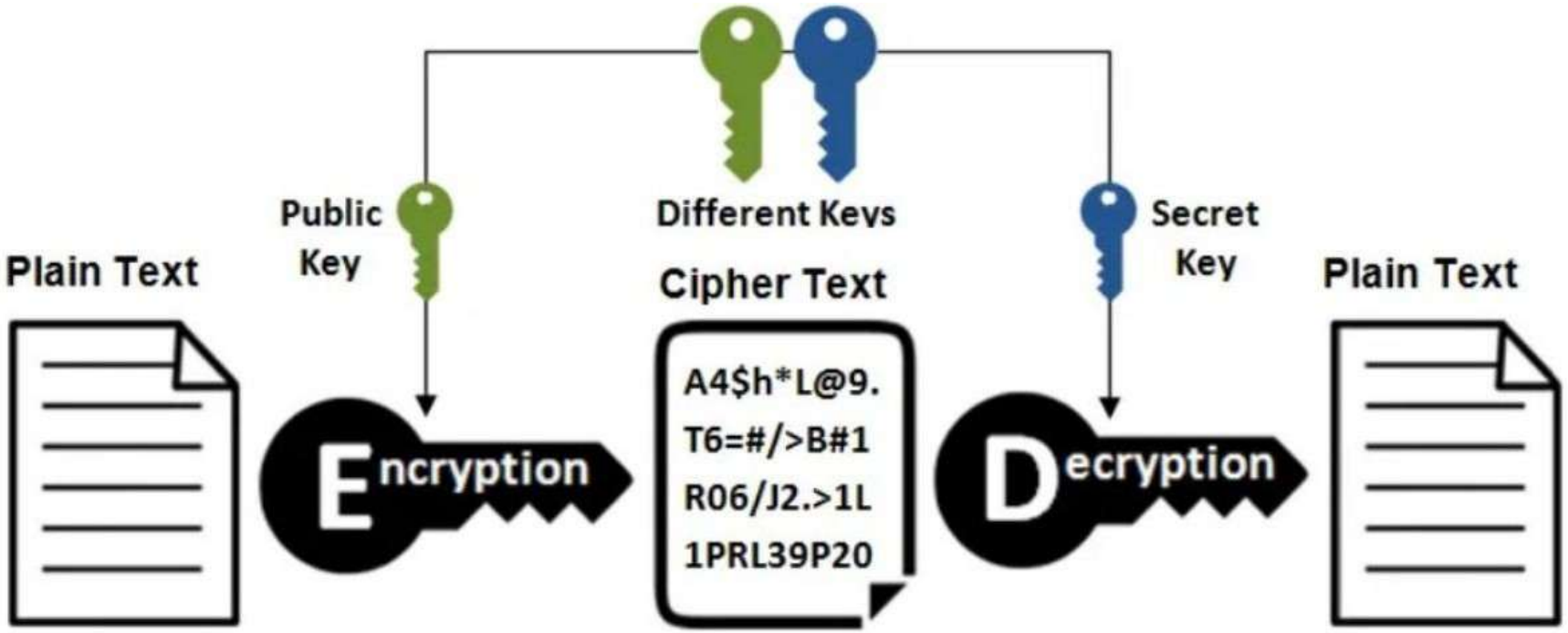
Symmetric Key Cryptography

An encryption system in which the sender and receiver of a message share a single, common key that is used to encrypt and decrypt the message. The most popular symmetric–key system is the Data Encryption Standard (DES)



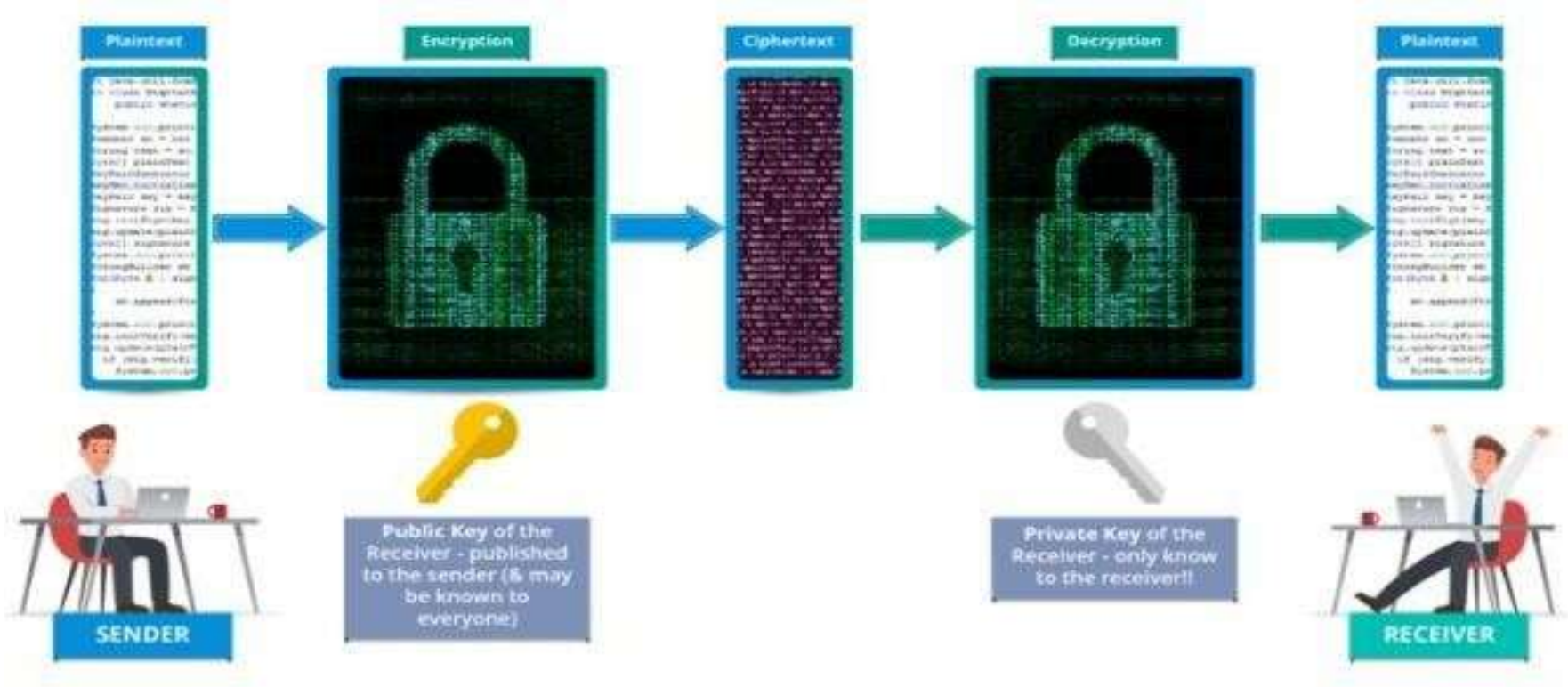
Asymmetric Key Encryption (or Public Key Cryptography)

The encryption process where different keys are used for encrypting and decrypting the information. Keys are different but are mathematically related, such that retrieving the plain text by decrypting ciphertext is feasible.



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Asymmetric Key Encryption (or Public Key Cryptography)



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Module 2:

Lecture 7: SYMMETRIC ENCRYPTION



Transposition Ciphers

In Cryptography, a transposition cipher is a method of encryption by which the positions held by units of plaintext (which are commonly characters or groups of characters) are shifted according to a regular system, so that the ciphertext constitutes a permutation of the plaintext.

1	2	3	4	5	6
M	E	E	T	M	E
A	F	T	E	R	P
A	R	T	Y		

4	2	1	6	3	5
T	E	M	E	E	M
E	F	A	P	T	R
Y	R	A		T	

Plain Text: MEET ME AFTER PARTY

Key Used: 421635

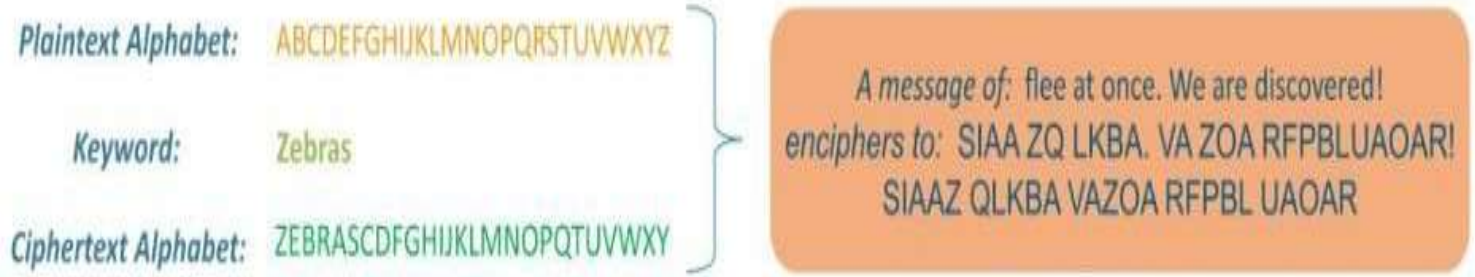
Cipher Text: TEMEEMEFAPTRYRAT



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Substitution Cipher

Method of encryption by which units of plaintext are replaced with ciphertext, according to a fixed system; the “units” may be single letters (the most common), pairs of letters, triplets of letters, mixtures of the above, and so forth.

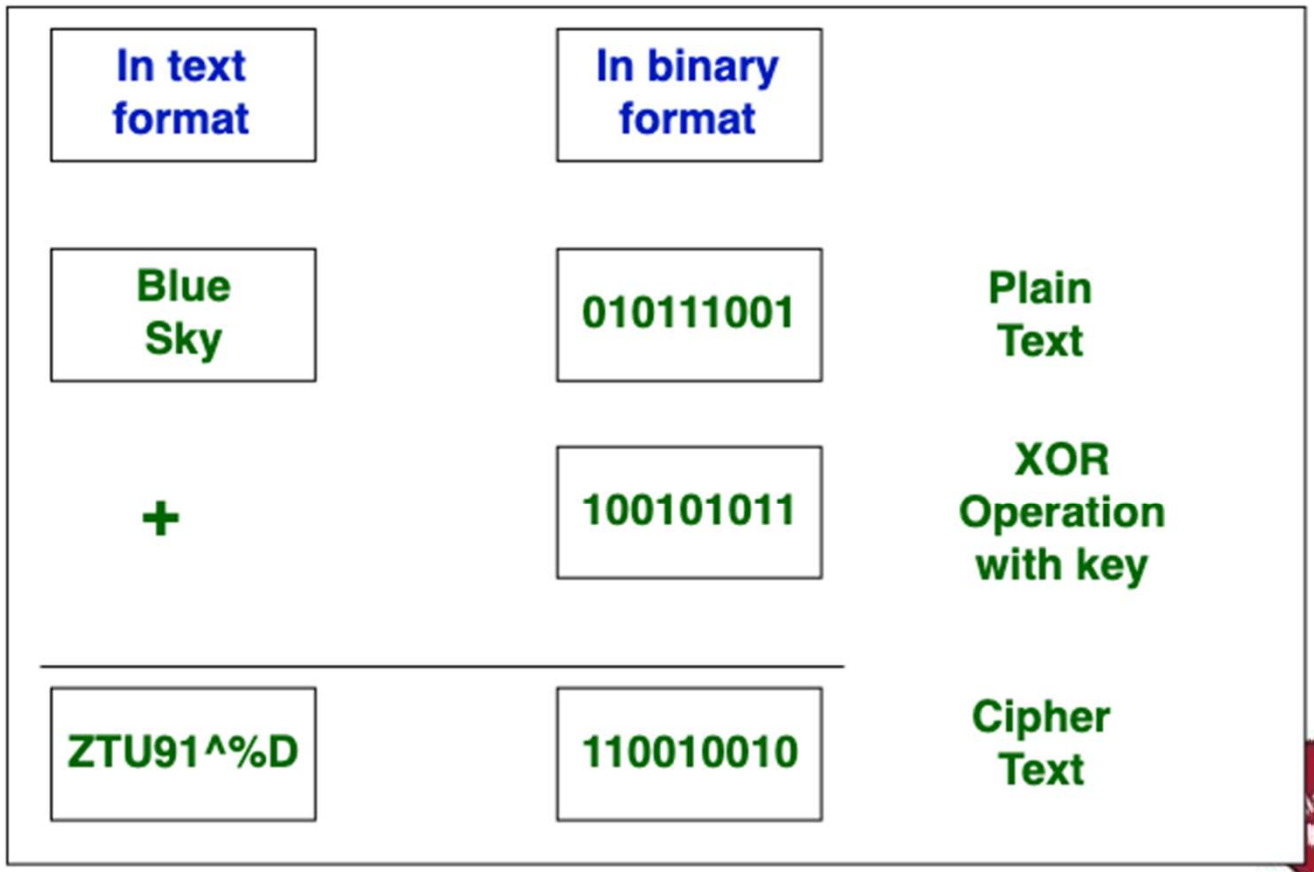


Stream Cipher

A stream cipher encrypts data one bit at a time rather than in fixed-size blocks in block cipher.

It generates a keystream that is combined with the plaintext to the produce ciphertext.

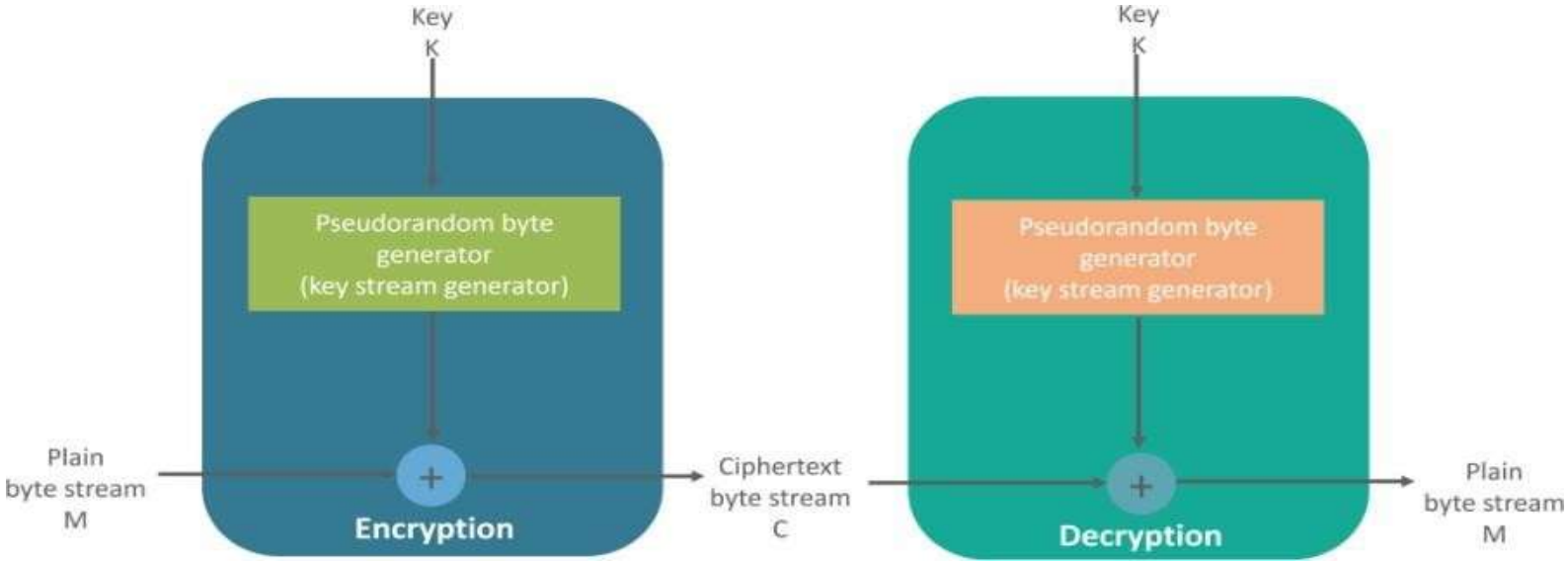
Stream ciphers are made for the scenarios where data needs to be encrypted in the continuous stream making them suitable for the real-time applications.



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Stream Cipher

Symmetric or secret-key encryption algorithm that encrypts a single bit at a time. With a Stream Cipher, the same plaintext bit or byte will encrypt to a different bit or byte every time it is encrypted.

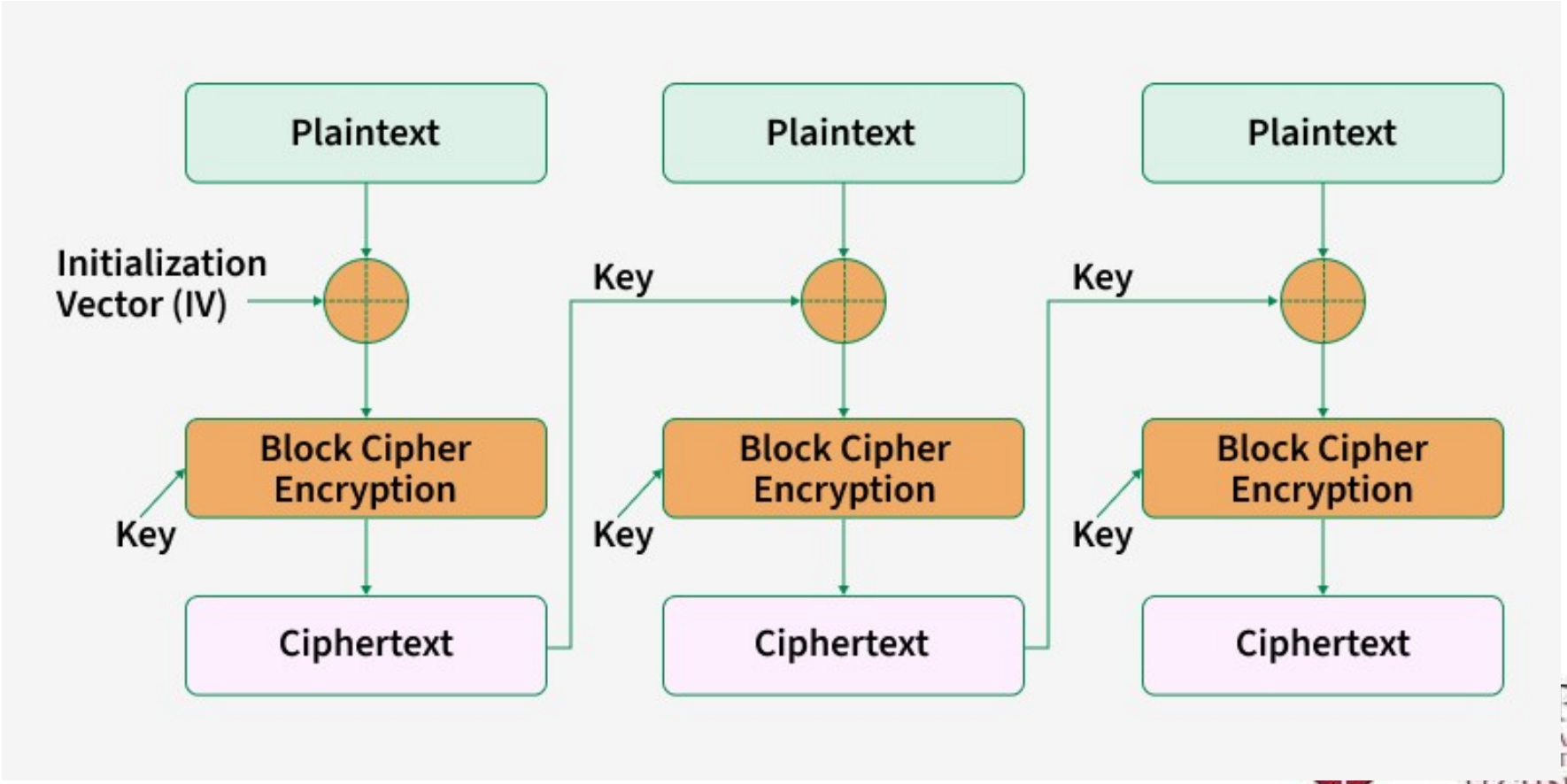


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Block Cipher

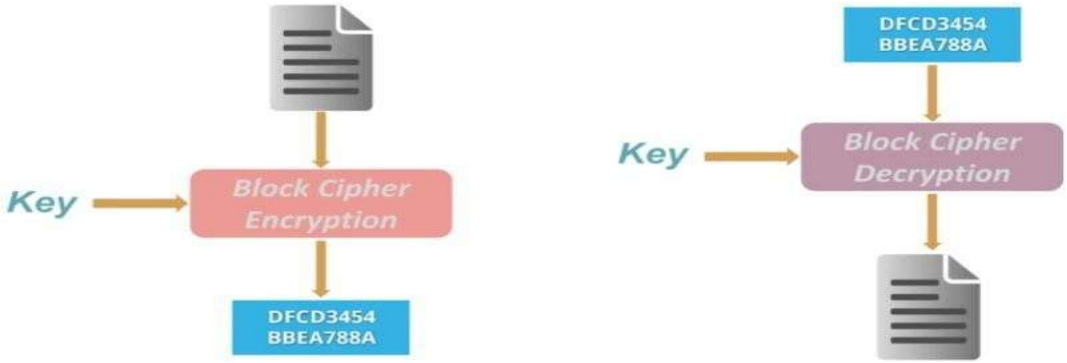
A block cipher encrypts data in fixed-size blocks usually 64 or 128 bits at a time. The encryption algorithm processes each block of data separately using the cryptographic key to transform the plaintext into the ciphertext.

Block ciphers function on complex mathematical computation and permutation to ensure that the data encrypted is safe. The choice of block size does not directly affect the strength of the encryption scheme.



Block Cipher

An encryption method that applies a deterministic algorithm along with a symmetric key to encrypt a block of text, rather than encrypting one bit at a time as in stream ciphers



Example: A common block cipher, AES, encrypts 128-bit blocks with a key of predetermined length: 128, 192, or 256 bits. Block ciphers are pseudorandom permutation (PRP) families that operate on the fixed size block of bits. PRPs are functions that cannot be differentiated from completely random permutations and thus, are considered reliable until proven unreliable.

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Difference Between Block Cipher and Stream Cipher

Block Cipher	Stream Cipher
Block Cipher Converts the plain text into cipher text by taking plain text's block at a time.	Stream Cipher Converts the plain text into cipher text by taking 1 bit plain text at a time.
Block cipher uses either 64 bits or more than 64 bits.	While stream cipher uses 8 bits.
The complexity of block cipher is simple.	While stream cipher is more complex.
Block cipher uses confusion as well as diffusion.	While stream cipher uses only confusion.
In block cipher, reverse encrypted text is hard.	While in-stream cipher, reverse encrypted text is easy.
The algorithm modes which are used in block cipher are ECB (Electronic Code Book) and CBC (Cipher Block Chaining).	The algorithm modes which are used in stream cipher are CFB (Cipher Feedback) and OFB (Output Feedback).
Block cipher works on transposition techniques like rail-fence technique, columnar transposition technique, etc.	While stream cipher works on substitution techniques like Caesar cipher, polygram substitution cipher, etc.
Block cipher is slow as compared to a stream cipher.	While stream cipher is fast in comparison to block cipher.
More secure than stream ciphers when the same key is used multiple times.	Less secure than block ciphers when the same key is used multiple times.
Key length is typically 128 or 256 bits.	Key length is typically 128 or 256 bits.
Operates on fixed-length blocks of data.	Encrypts data one bit at a time.

Module 2:

Lecture 10: DES-Block-SYMMETRIC ENCRYPTION



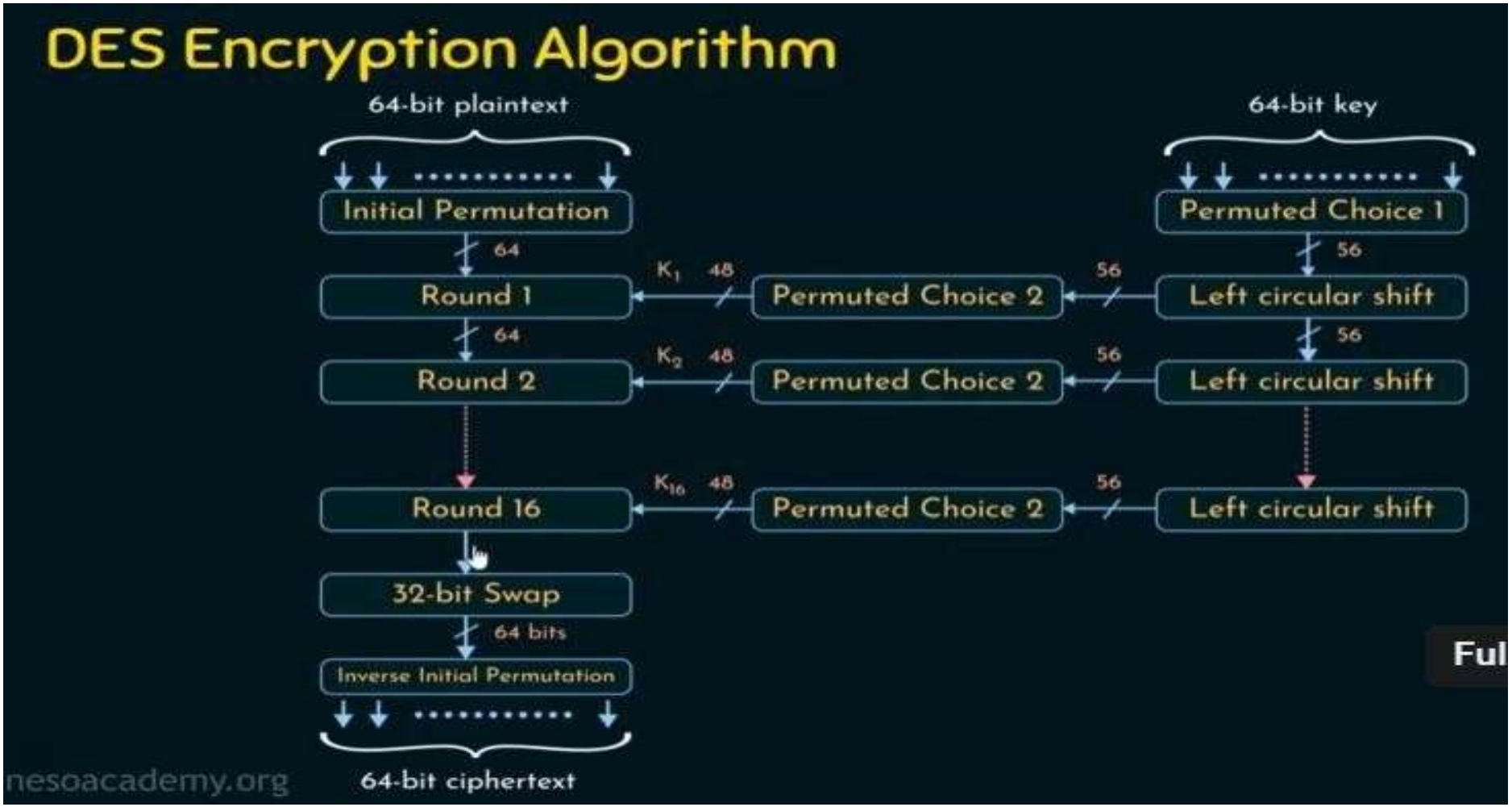
Data Encryption Standard

- The Data Encryption Standard (**DES**) is a symmetric-key block cipher.
- It encrypts and decrypts data in 64-bit blocks using a single 56-bit key.
- IBM developed DES in the 1970s. Because its short key makes it vulnerable to modern computing power, it was historically upgraded to Triple DES (3DES) and is now largely replaced by AES.



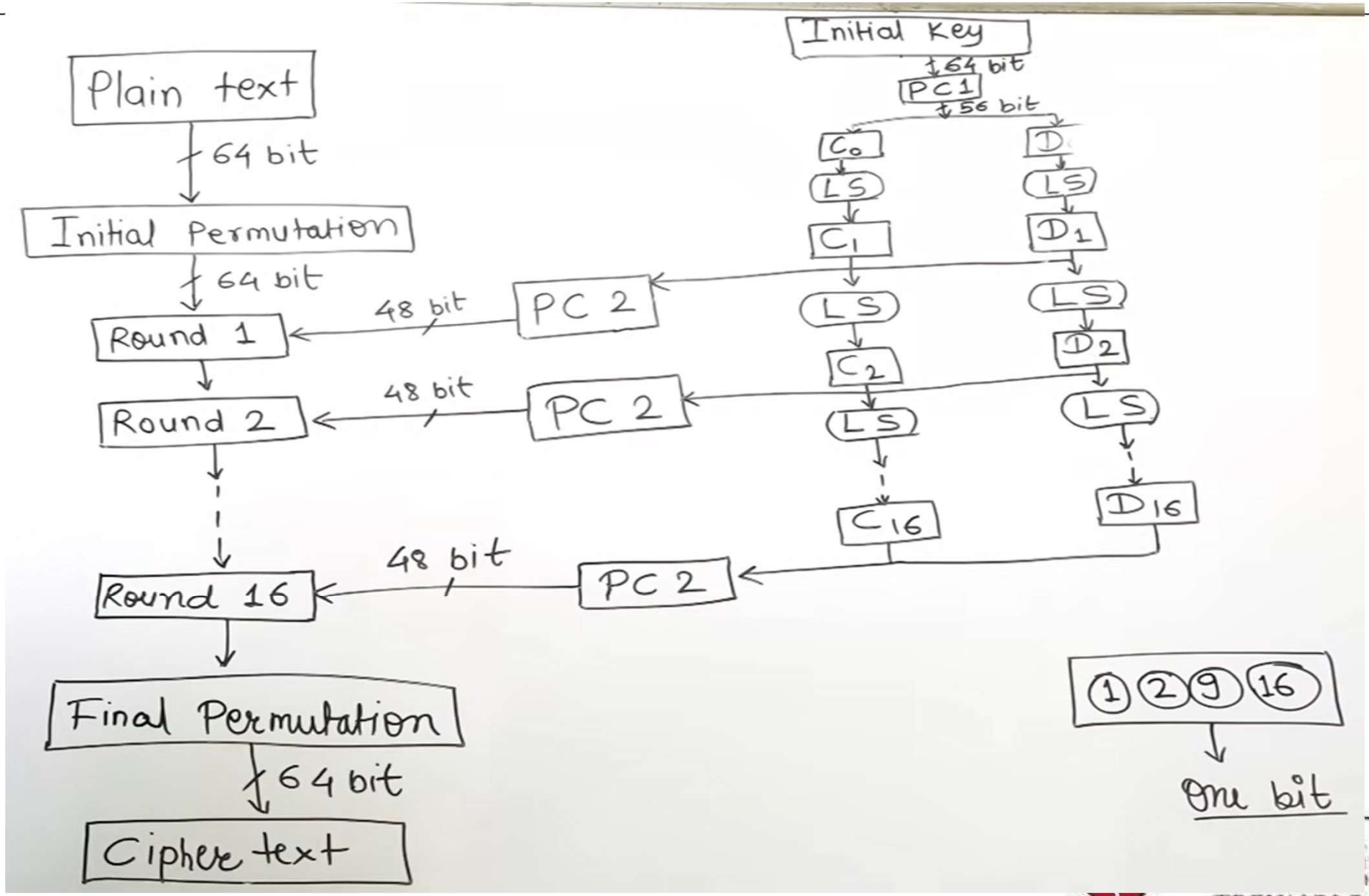
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Data Encryption Standard



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Data Encryption Standard



How DES Works

DES uses a **Feistel cipher** structure. This means it breaks data into two halves and runs multiple cycles of mixing and replacing.

1.Initial Permutation (IP): The algorithm takes a 64-bit block of plain text and rearranges the bits in a fixed order.

2.16 Rounds of Processing: The block is split into left and right 32-bit halves. Over 16 rounds, the right half undergoes complex scrambling. It is expanded to 48 bits, mixed with a round key (XORed), shrunk using lookup tables (S-boxes), and permuted.

3.32-Bit Swap: After the 16 rounds, the left and right halves are swapped.

4.Inverse Initial Permutation (IP^{-1}): Finally, a reverse permutation is applied to yield the 64-bit ciphertext.

4.Key Scheduling: Even though the main DES key is 64 bits long, 8 of those bits are used as check bits, leaving an effective key length of 56 bits. In every round, a unique 48-bit subkey is created by performing circular left shifts and specific permutations on the main 56-bit key.

Module 2:

Lecture 11: AES-Block-SYMMETRIC ENCRYPTION



Advanced Encryption Standard(AES)

Advanced Encryption Standard (AES) is a specification for the encryption of electronic data established by the U.S National Institute of Standards and Technology (NIST) in 2001. AES is widely used today as it is a much stronger than DES and triple DES despite being harder to implement.

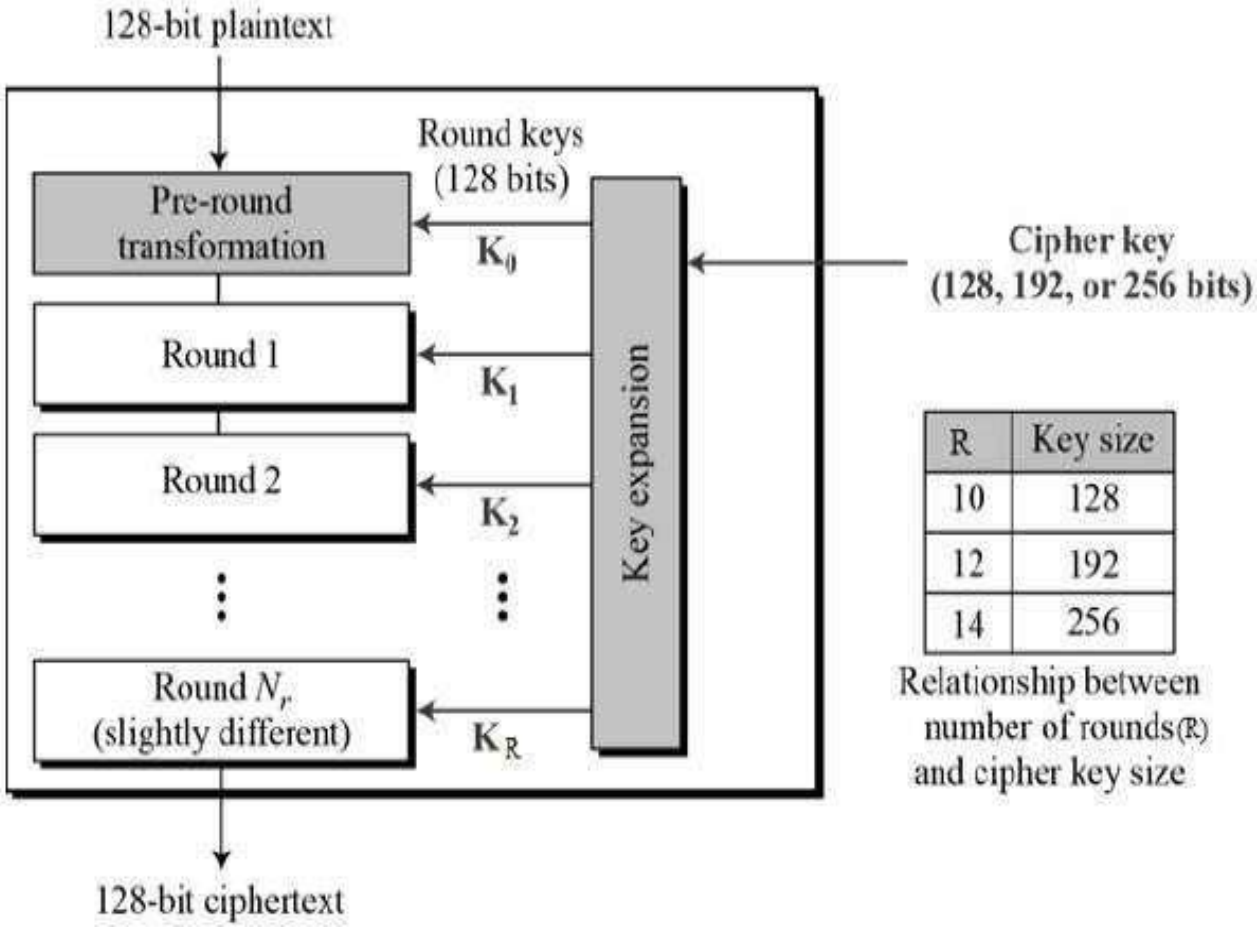
- AES is a block cipher.
- The key size can be 128/192/256 bits.
- Encrypts data in blocks of 128 bits each

The number of rounds depends on the key length as follows :

- 128 bit key – 10 rounds
- 192 bit key – 12 rounds
- 256 bit key – 14 rounds

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Advanced Encryption Standard(AES)



R	Key size
10	128
12	192
14	256

Relationship between number of rounds(R) and cipher key size

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Advanced Encryption Standard(AES)

- AES considers each block as a 16 byte (4 byte x 4 byte = 128) grid in a column major arrangement.

```
[ b0 | b4 | b8 | b12 |
| b1 | b5 | b9 | b13 |
| b2 | b6 | b10| b14 |
| b3 | b7 | b11| b15 ]
```

- Each round comprises of 4 steps :
  - SubBytes: Performs substitution
  - ShiftRows: Performs permutation
  - MixColumns: Performs permutation. Last round doesn't have MixColumn.
  - Add Round Key

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Advanced Encryption Standard(AES)

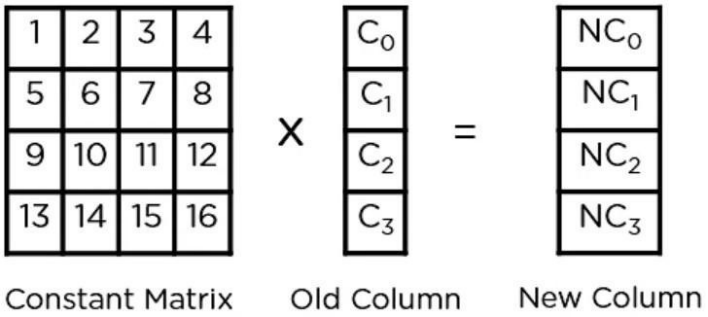
- SubBytes: In this step each byte is substituted by another byte. Its performed using a lookup table also called the S-box. This substitution is done in a way that a byte is never substituted by itself and also not substituted by another byte which is a compliment of the current byte. The result of this step is a 16 byte (4 x 4) matrix.
- ShiftRows :
  - This step is just as it sounds. Each row is shifted a particular number of times.
  - The first row is not shifted
  - The second row is shifted once to the left.
  - The third row is shifted twice to the left.
  - The fourth row is shifted thrice to the left.



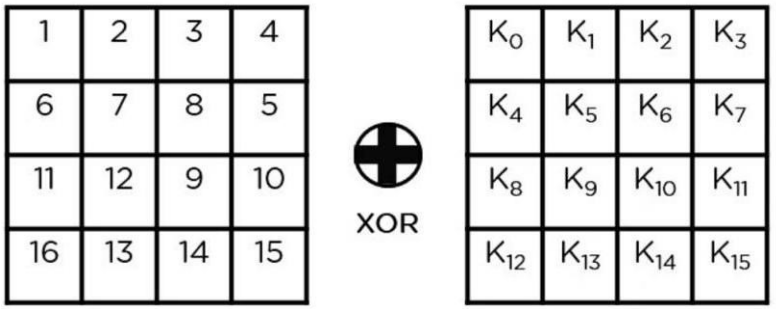
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Advanced Encryption Standard(AES)

- MixColumns : This step is basically a matrix multiplication. Each column is multiplied with a specific matrix and thus the position of each byte in the column is changed as a result.



- Add Round Key: Now the resultant output of the previous stage is XOR-ed with the corresponding round key. Here, the 16 bytes is not considered as a grid but just as 128 bits of data.



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AES and DES Comparision

Parameters	AES	DES
key	128, 192, or 256 bits	56 bits
Rounds	(10, 12, or 14)depends upon key	16
Structure	The structure is based on a substitution-permutation network.	The structure is based on a Feistel network.
Security	AES is more secure than the DES cipher	DES can be broken easily due to current high computation power.
Rounds	The rounds in AES are: Byte Substitution, Shift Row, Mix Column and Key Addition	The rounds in DES are: Expansion, XOR operation with round key, Substitution and Permutation
Plaintext size.	AES can encrypt 128 bits of plaintext.	DES can encrypt 64 bits of plaintext.

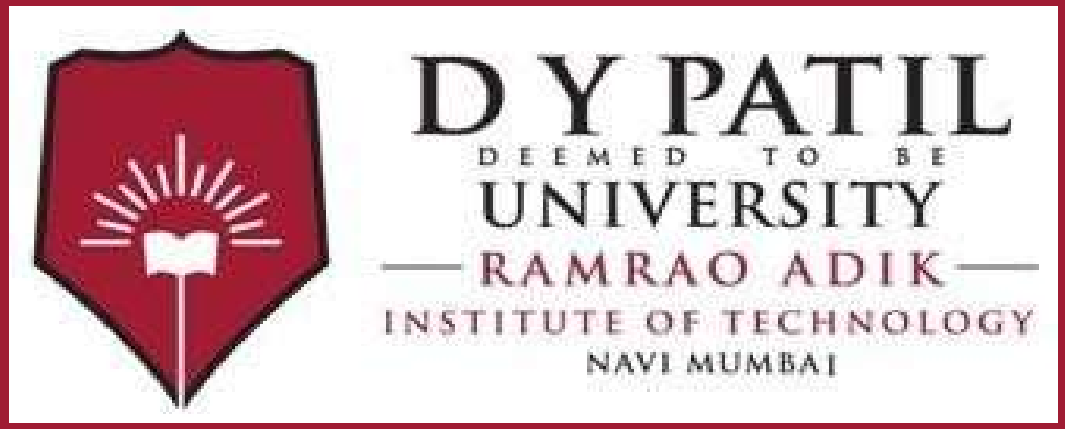


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AES and DES Comparision:

Application	web browsers, email clients, VPNs, wireless networks, and cloud computing	DES is still used in some legacy systems that cannot support newer ciphers
Use:	AES is also compatible with other cryptographic protocols such as SSL, TLS, IPsec, and SSH	Not used at protocol level
	It is efficient with both hardware and software.	It is efficient with hardware.

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Thank You